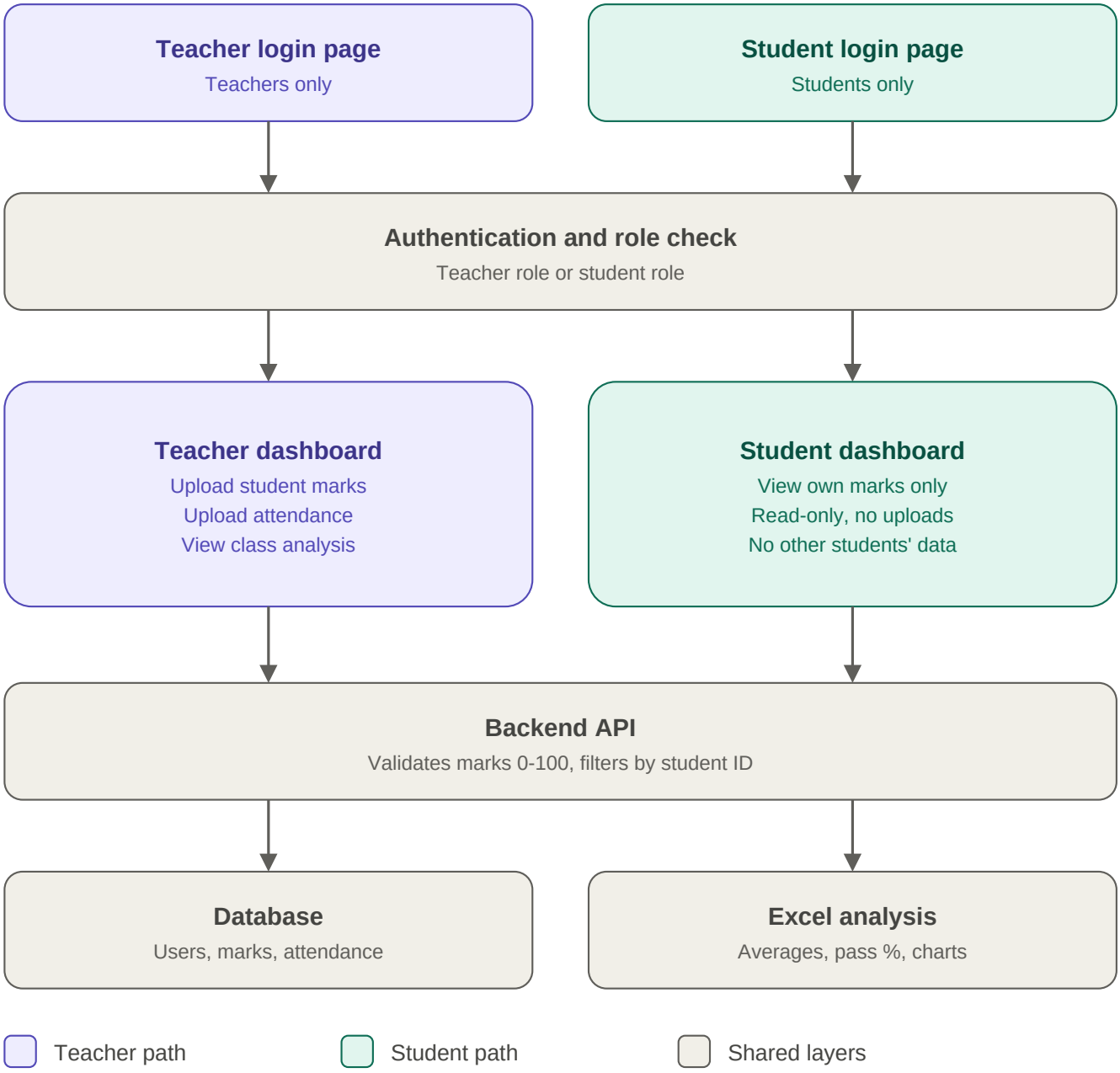
Student Performance Portal

System architecture - separate teacher and student login pages



Based on the Student Performance Analysis README (sample data, practice purposes only).

How the access rules work

- **Two separate login pages.** Each page only accepts accounts with its own role, so a student cannot sign in through the teacher page.
- **Role check on every request.** Logging in is not enough. The backend checks the role each time, so a student cannot reach the upload endpoints even by typing the URL.
- **Teacher dashboard.** Teachers upload marks and attendance (by form or by uploading an Excel/CSV file) and can view class-level analysis such as subject averages and pass percentage.
- **Student dashboard.** Read-only. The API returns only rows where the student ID matches the logged-in student, so no one can see another student's marks.
- **Backend validation.** Marks must be numeric and between 0 and 100, with no duplicate records. The pass rule (at least 40 in every subject) is applied here, so Result is calculated, never typed by hand.

Suggested database tables

Table	Columns
users	id, email, password hash, role (teacher or student), student ID (students only)
marks	student ID, subject, marks, uploaded by, date
attendance	student ID, date or month, status or percentage, uploaded by

Excel analysis

The original Excel workbook stays in the design. The database exports to it for the dashboard, subject averages, pass percentage and charts described in the project README.